

CS & IT ENGINEERING



Operating System

Process Management

Lecture - 2

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Recap of Previous Lecture



doubt?

Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block



Topics to be Covered



Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers

CPU





Topic : Process States

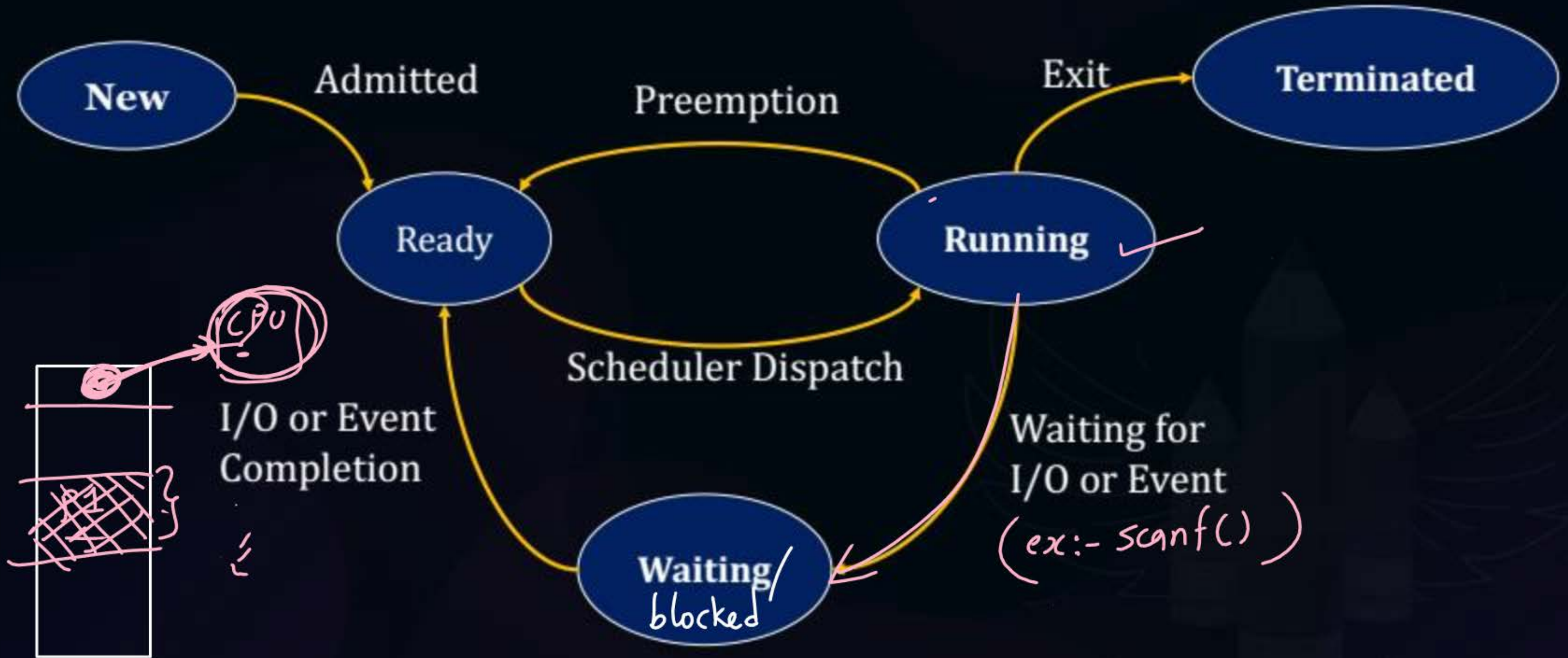


↓
defined based on what process is doing currently.





Topic : Process States





Topic : Process States

New:-

All installed processes are known to be in new state, when they are not running.

Ready:-

All processes which are waiting to run on CPU are known to be in ready state

Running:-

A process which is running on CPU has its state as running

Terminated:-

A completed process has its state as terminated

Blocked:-

All processes which are waiting for any I/O or event



Topic : Process States



New To Ready: When process is admitted by OS

Ready to Running: When a process is dispatched to CPU

Running to Terminated: When a process is completed

Running to Blocked: When a process goes for IO or event

Running to Ready: When a process is preempted

Blocked to Ready: When a process completes IO or event



Topic : Process States



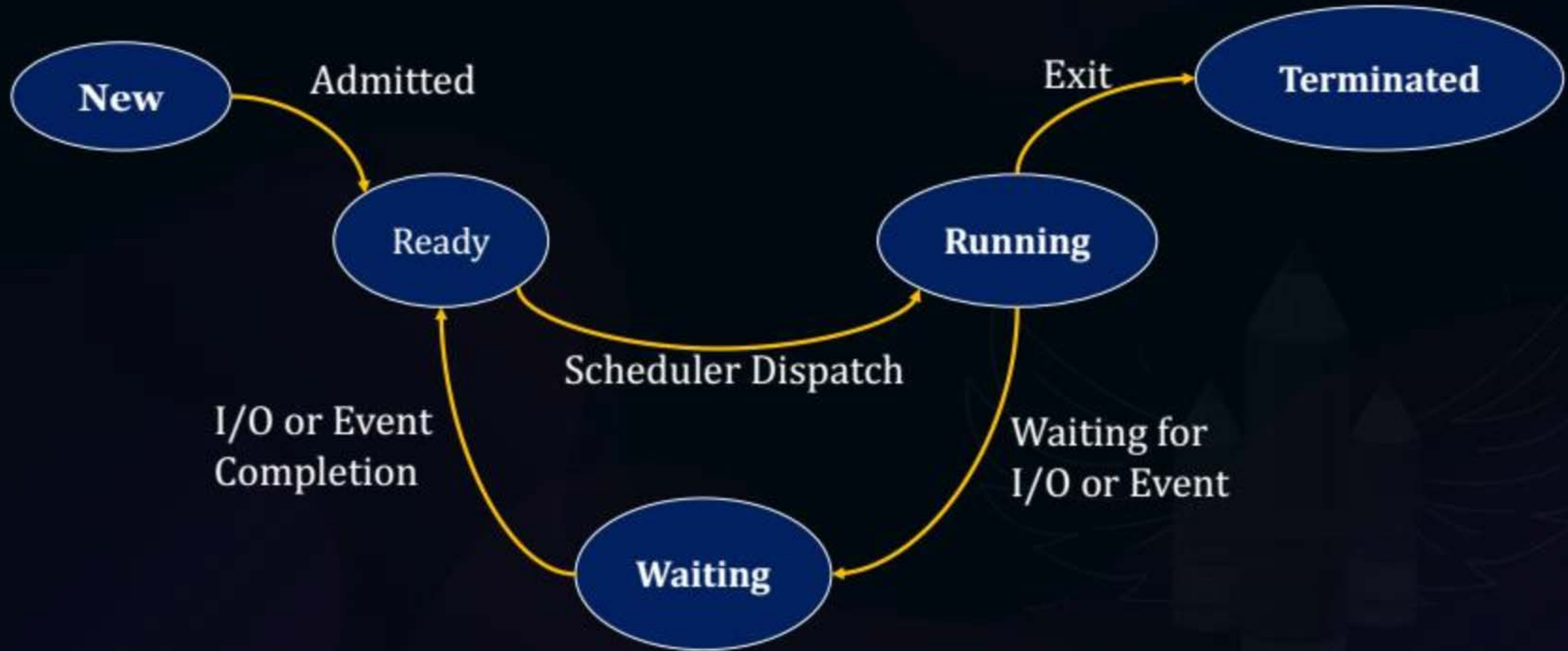
❑ 2 Transitions are voluntary:

- Running to Terminated
- Running to Blocked

Rest all are initiated by OS



Topic : Non-preemptive Process States





Topic : CPU vs IO Bound Process



(CPU extensive)

CPU Bound: If the process is intensive in terms of CPU operations

IO Bound: If the process is intensive in terms of IO operations





Topic : Process Scheduling

Needed Because?

to achieve best possible performance



Topic : Scheduling Queues

- Job Queue → list in which all new state processes are kept
- Ready Queue → list in which all ready state — || —
- Device Queue → list in which all blocked state processes are kept, which are waiting for a device.



multiple device queues are present.
one for each device.

Dispatcher :-

program of OS which performs context switch.



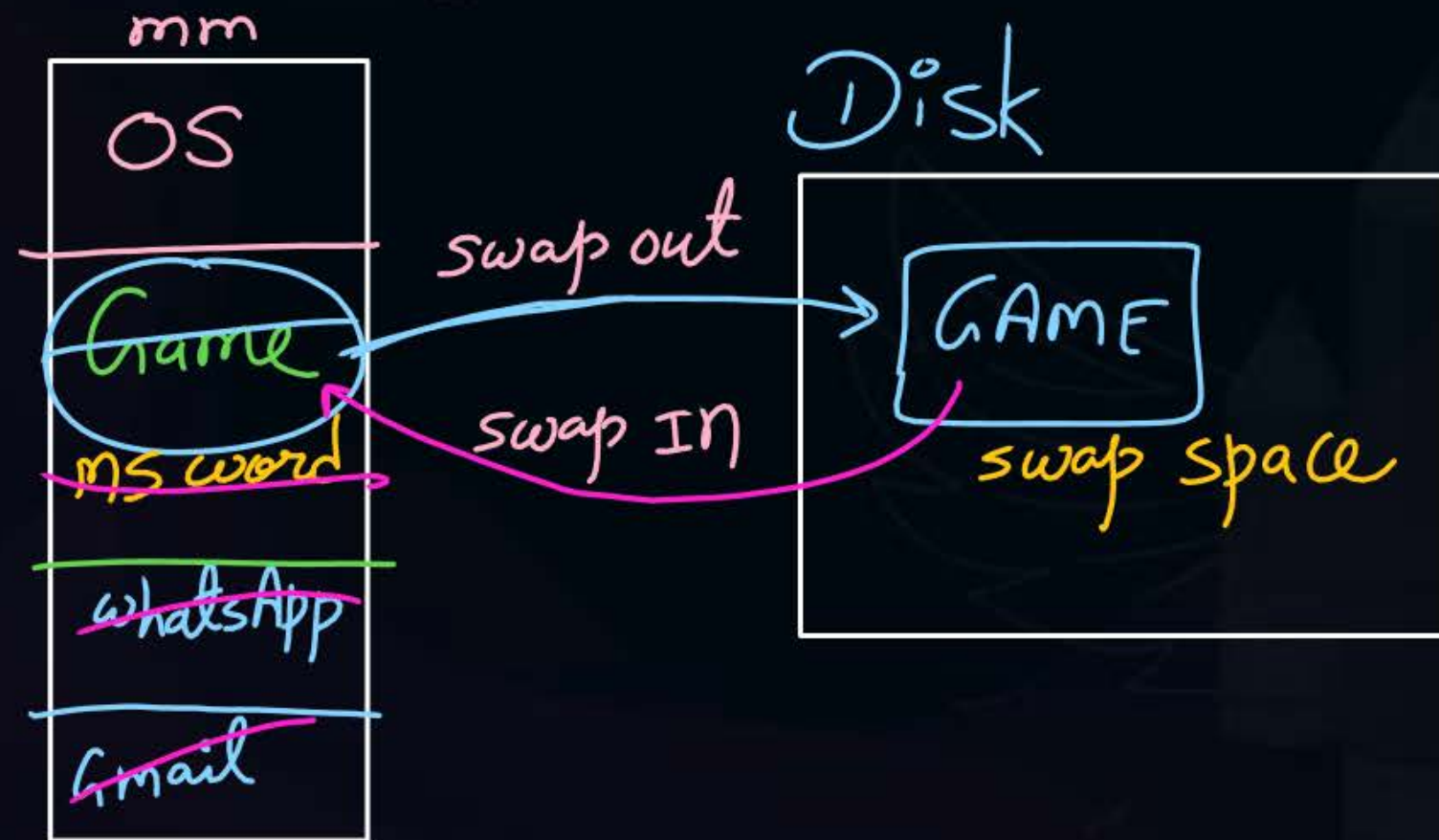
Topic : Types of Schedulers

→ Programs in OS

- Long-Term Scheduler (Job) → It brings processes from new state to ready state
- Short-Term Scheduler (CPU) → It selects one of the ready state processes to run next on CPU.
- Mid-Term Scheduler (Medium-term)

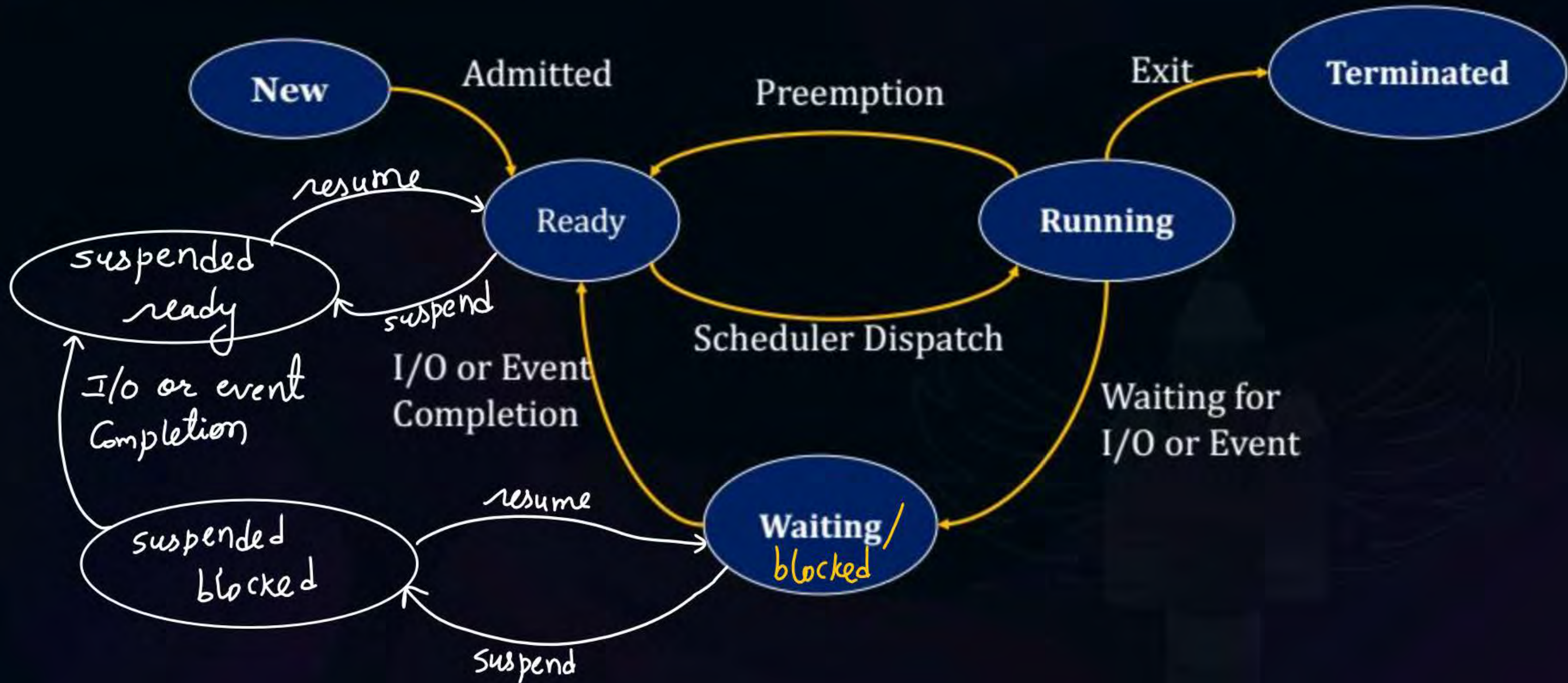
⇓
It performs swapping of processes

If swapping is performed based on priority of processes then it is called as rolling.





Topic : Process States



#Q. Which of the following scheduler reduces the degree of multiprogramming?

- A** Short-Term
- B** Long-Term
- C** Mid-Term
- D** Long-Term and Mid-Term both





2 mins Summary

Topic

Process states

Topic

Process State Transition

Topic

Process Scheduling

Topic

Types of Schedulers



Happy Learning

THANK - YOU

